

TOWARDS REVERSIBLE MODEL MERGING FOR LOW-RANK WEIGHTS

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ABSTRACT

Model merging consolidates multiple fine-tuned models into a single set of weights intended to perform well across all source tasks. Although prior work has established that merging can approximate the per-task accuracy of individually fine-tuned models, it has systematically overlooked the regime that now predominates in practice—the setting in which models are stored in low-rank form, whether through low-rank adaptation (LoRA) or post-training singular value decomposition (SVD). We show that applying conventional merging methods to such low-rank weights precipitates severe degradation of the merged model, and we trace this failure to two coupled mechanisms: task interference amplified by the limited representational capacity of compressed updates, and the destruction of the coupling between the low-rank factors that only their product renders meaningful. Because this collapse persists across compression schemes, ranks, and merging algorithms, we identify it as intrinsic to the single-model objective rather than as a tuning artifact, a diagnosis that motivates a fundamentally different formulation. Instead of collapsing all adapters into one set of weights, we construct a compact basis—equivalent in storage cost to retaining only a few models—from which every original task-specific model can be recovered by linear combination. Merging thereby becomes the construction of a reconstruction-capable model space rather than the production of a single merged model, a reformulation that makes it possible to “revert” to any individual model on demand and that squarely confronts the fact that no single merged model can consistently match a model specialized for its task. Building on this principle, we introduce Reversible Model Merging (RMM), an efficient, data-free, and flexible method that admits a closed-form solution for jointly selecting the optimal basis of model weights together with the task-specific coefficients that govern their linear combination. Extensive experiments spanning diverse datasets, backbone architectures that comprise both encoder and decoder language models as well as vision transformers, and both LoRA and PT-SVD compression regimes demonstrate that RMM consistently and substantially outperforms existing merging approaches, preserving the accuracy of low-rank compressed models by a wide margin while retaining only a fraction of the storage required to keep every adapter.

1 INTRODUCTION

Parameter-efficient fine-tuning methods, foremost among them Low-Rank Adapters (LoRA) (Hu et al., 2021), have become a cornerstone of adapting large models to diverse downstream tasks without incurring the full cost of retraining. In parallel, post-hoc compression techniques—notably post-training singular value decomposition (SVD) truncation—are now routinely deployed to curb storage and inference costs, contributing a complementary axis of efficiency (Ryu et al., 2023; Ping et al., 2024; Wang et al., 2025b;a). Against this backdrop, model merging has emerged as a compelling paradigm for consolidating the knowledge of many task-specific models into a single unified set of weights. A rapidly expanding body of work, spanning everything from simple parameter interpolation to optimization-based schemes, has established that merged models can perform competitively across source tasks without access to the original training data (Matena & Raffel, 2022; Yang et al., 2024; Yadav et al., 2023; Wei et al., 2025). This literature has nevertheless developed almost entirely on full-rank models, leaving the interaction between merging and low-rank compres-

sion—precisely the regime in which fine-tuned models are now predominantly stored—essentially unexamined. We close this gap directly, and in doing so uncover a failure mode that reshapes how merging must be formulated in the compressed setting.

Despite this progress, a fundamental limitation remains unaddressed: even the most sophisticated merging strategies consistently fall short of the accuracy of the original, individually fine-tuned models. This shortfall widens sharply once the models being merged have already been compressed into low-rank form, whether through LoRA fine-tuning or post-training SVD truncation. The cause is structural rather than incidental. In enabling a single model to serve many tasks, merging necessarily collapses distinct task-specific representations into one shared parameter space, and the interference induced by this collapse erodes the specialized adaptations that each model acquired during fine-tuning (Gargiulo et al., 2025). We argue that this tension between consolidation and fidelity is not an artifact of imperfect algorithms but is inherent to the single-model formulation of merging itself—a distinction that, as we demonstrate, determines where an effective solution must be sought and rules out the incremental refinement of existing coefficient-tuning schemes.

Although low-rank adapters and SVD-truncated models are individually compact, their proliferation introduces a distinct scalability bottleneck at the level of the collection. In realistic multi-task, federated, and continual-learning deployments, practitioners routinely accumulate dozens or even hundreds of task-specific low-rank adapters, each demanding separate storage and coordinated management. Retaining all of them forfeits precisely the storage savings that low-rank parameterization was designed to deliver, and it renders deployment progressively more cumbersome as the collection grows. This state of affairs establishes a concrete and pressing need for merging strategies that operate directly on low-rank representations—not merely to consolidate storage and eliminate redundancy, but to enable the efficient sharing, deployment, and transfer of compressed models across tasks. Meeting this need is the central objective of the present work.

To ground this need in evidence, we begin by characterizing systematically how existing merging techniques behave when applied to low-rank compressed models. Our analysis centers on deltas—the weight differences between each fine-tuned model and the shared pre-trained initialization—which conventional merging strategies attempt to combine into a single set of parameters under an implicit assumption of linear compatibility. We establish empirically that merging low-rank deltas in this manner precipitates catastrophic degradation of task performance. The failure is attributable to two coupled factors: the limited expressive capacity inherent to low-rank representations, and the misalignment of task-specific subspaces that, having been optimized independently, interfere destructively when superimposed. Because the collapse persists uniformly across compression schemes, ranks, and merging algorithms, these findings identify the difficulty not as a tuning artifact but as a systematic incompatibility between the single-model merging objective and compressed representations—an incompatibility that no reweighting within that objective can resolve.

This diagnosis compels us to reconsider the objective of merging itself: *What if merging need not terminate in a single final model? What if the problem were relaxed to permit retaining more than one component? What if we preserved the ability to reconstruct each original model on demand?* Pursuing these questions, we replace a single collapsed set of weights with a compact set of weights that functions as a shared basis—one from which every original model can be recovered by linear combination. This reframing opens a largely unexplored design space: rather than compressing all information into one model, we retain complementary representational directions across a small set of basis models, and thereby enable post-merging “reversion” to the original adapters. In doing so, our approach overturns the prevailing assumption that merging must yield a single model and recasts merging as the construction of a reconstruction-capable model space—a shift in objective that, as the remainder of the paper shows, is both theoretically principled and empirically decisive.

Building on this reconstruction-driven perspective, we develop a theoretical framework that characterizes the optimal basis for minimizing reconstruction error and instantiate it as our method, Reversible Model Merging (RMM). RMM recasts merging as a basis-selection problem whose objective is to identify a compact set of task vectors that supports accurate reconstruction of all original models. Within this framework, RMM admits a closed-form solution that simultaneously determines a provably optimal set of basis task vectors and the task-specific coefficients that recover each individual adaptation by linear combination, without any access to task data or iterative optimization. We summarize our contributions as follows.

- **Diagnosis of merging under compression.** We identify and empirically characterize the catastrophic failure of conventional merging on low-rank compressed models, and we trace it to two concrete mechanisms—amplified task interference under limited representational capacity and the broken coupling between the low-rank factors—thereby establishing that the failure is intrinsic to the single-model formulation rather than attributable to any particular algorithm.
- **Reversible reformulation with a closed-form solution.** We reformulate merging as the construction of a reconstruction-capable model space and derive, via Theorem 1, a closed-form, data-free basis-selection solution that jointly yields the provably optimal orthonormal basis and the task-specific reconstruction coefficients, eliminating the iterative optimization and data access on which competing methods depend.
- **Comprehensive empirical validation.** Across a broad range of datasets, backbone architectures, and compression regimes, we demonstrate that RMM surpasses conventional merging strategies by a substantial and uniform margin—without a single exception across compression schemes, ranks, and tasks—while preserving the efficiency benefits of low-rank parameterization, and we show that a single tunable hyperparameter, the number of retained components, exposes a controllable storage–fidelity trade-off that scales favorably to large model collections.

2 RELATED WORK

Model merging has established itself as a powerful strategy for consolidating knowledge from multiple fine-tuned models. Existing methods fall into two broad families: data-free parameter-space manipulations, and optimization-based approaches that leverage data or auxiliary objectives. We review representative techniques from both lines below, organizing the discussion around how each confronts the interference that arises when task-specific updates are superimposed—the very phenomenon that, as we later demonstrate, compression amplifies to the point of collapse. This organizing lens also clarifies what unites the entire literature and separates it from our approach: every method surveyed here retains the single-model assumption that RMM discards.

Task Arithmetic: The foundational abstraction underlying modern merging is task arithmetic (Ilharco et al., 2023), which defines a delta as the difference between the parameters of a fine-tuned model and its pre-trained initialization. Its central insight is that such deltas behave as modular, composable building blocks: they can be added to a base model to impart new abilities, subtracted to suppress undesired behaviors, or scaled and combined to blend multiple capabilities. This linear view of weight-space composition supplies the conceptual foundation on which the subsequent interference-aware methods—and, as we show, our own reversible formulation—are built, since it is precisely the linearity of delta composition that RMM exploits to reconstruct each model from a shared basis.

DARE: Although simple delta addition can be effective, it degrades markedly when the merged models carry conflicting or overlapping parameter updates. To counter this, Yu et al. (2024) introduced DARE (Drop and Rescale), a method built on aggressive randomized pruning. DARE discards the vast majority of parameter changes—typically 90–99%—on the premise that most updates are inessential and may actively interfere with the knowledge contributed by other models, and it subsequently rescales the surviving updates to preserve the overall magnitude of the modification. The net effect is to retain the core task-specific components of each delta while suppressing the interference that would otherwise accompany dense superposition—an effect that, as our diagnosis in Section 4 makes clear, becomes untenable once the representational capacity available for pruning has already been exhausted by compression.

TIES: Where DARE resolves conflicts stochastically, TIES-merging supplies a deterministic three-phase procedure—Trim, Elect Sign, and Merge—that resolves interference across models systematically (Yadav et al., 2023). It first trims each delta to its largest-magnitude entries, retaining the most influential fine-tuning updates; it then elects, at each parameter position, a majority sign across models and discards the contributions that oppose it; and it finally averages the sign-consistent parameters into the merged representation. This consensus-driven pipeline curtails destructive interference and yields a more stable merged model. Motivated by the same interference-reduction objective, EMR-Merging (Elect, Mask, and Rescale) elects a single unified model and attaches task-specific binary masks and rescalers, derived from each individual model, that steer the unified model toward each task at inference time (Huang et al., 2024). Notably, EMR-Merging retains lightweight per-

task information at inference—a strategy that RMM generalizes and renders optimal by replacing heuristic masks with a provably optimal reconstruction basis.

Other Merging Methods: A complementary family moves beyond purely parameter-space manipulation and instead relies on explicit optimization or access to a small dataset. Model soups (Wortsman et al., 2022) refine the task-arithmetic merging weights by minimizing validation loss, steering the interpolation toward better generalization, whereas Fisher-weighted averaging (Matena & Raffel, 2022) injects second-order information into the merge, up-weighting the parameters estimated to be most critical for task performance. Others cast merging directly as an optimization problem: RegMean (Jin et al., 2023) and AdaMerging (Yang et al., 2024) adaptively search for task-dependent merging coefficients by solving small-scale objectives—the former on the discrepancy between merged- and individual-model predictions, the latter on the entropy of predictions over unlabeled data. More recently, Wei et al. (2025) formulate merging as the minimization of the loss difference between the merged and individual fine-tuned models and derive a tractable surrogate through a first-order Taylor expansion. These optimization-based methods can attain stronger task-specific performance, yet they characteristically require labeled or unlabeled data, additional optimization, or both—prerequisites that sharply constrain their applicability precisely in the data-free, retraining-free regime that motivates our work. RMM, by contrast, retains the fully data-free character of the parameter-space family while dispensing with the single-model assumption that every one of these methods shares, and it does so through a closed-form solution that requires neither validation data nor iterative search.

3 PRELIMINARIES

We formalize the problem of consolidating a collection of fine-tuned models into a single unified model through merging. Let $\{\theta_1, \dots, \theta_n\}$ denote n models fine-tuned on n distinct tasks. A merging algorithm $\mathcal{M}$ maps these models to a single set of weights $\theta_{\mathcal{M}} = \mathcal{M}(\{\theta_i\}_{i=1}^n)$. The objective is to determine a merging procedure $\mathcal{M}$ that maximizes the average validation performance across all n tasks:

$$\max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\theta_{\mathcal{M}}, \mathcal{D}_i), \quad (1)$$

where $\mathcal{D}_i$ denotes a held-out validation set for task i and Perf denotes any suitable performance metric (e.g., accuracy or F1 score). Following standard practice, we assume that all fine-tuned models share a common pretrained initialization θ_{pre} . For any layer l , the update (or delta) of model i relative to this base is defined as $\Delta_i^l = \theta_i^l - \theta_{\text{pre}}^l$, where $\Delta_i^l \in \mathbb{R}^{m \times d}$ denotes the weight difference for a layer of dimensions $m \times d$. The delta thus isolates the task-specific adaptation from the shared initialization and constitutes the object on which all merging operations act.

3.1 LOW-RANK COMPRESSED MODELS

To curtail storage together with training and inference costs, fine-tuned models are routinely compressed into low-rank form. We consider the two most widely adopted compression strategies, which together account for the overwhelming majority of low-rank models encountered in practice:

LoRA fine-tuning: During training, each layer is augmented with trainable low-rank factors $\mathbf{A} \in \mathbb{R}^{m \times r}$ and $\mathbf{B} \in \mathbb{R}^{r \times d}$, with $r \ll \min(m, d)$. The pretrained weights are held frozen and optimization is confined to $\mathbf{A}$ and $\mathbf{B}$, so that the effective update is, by construction, constrained to a rank- r subspace throughout training.

Post-training SVD truncation (PT-SVD): After standard full fine-tuning, the update Δ_i^l is decomposed by SVD and truncated to its top- r singular components, which yields the low-rank approximation $\hat{\Delta}_i^l = \mathbf{A}_i^l \mathbf{B}_i^l$ with $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$, where $r \ll \min(m, d)$. In contrast to LoRA, PT-SVD imposes the low-rank constraint post hoc rather than during optimization, so that the two regimes together span both the training-time and post-training routes to low-rank storage.

In both regimes the fine-tuned update for layer l of model i therefore admits the common low-rank factorization $\hat{\Delta}_i^l = \mathbf{A}_i^l \mathbf{B}_i^l$, with $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$, $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$, and r the compression rank. This shared factorization is what allows us to treat LoRA and PT-SVD uniformly. Merging under compression

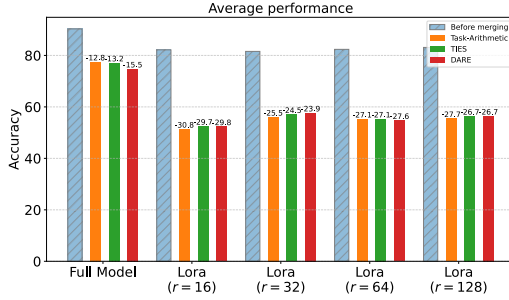


Figure 1: Average task performance of individual models (before merging) contrasted with merged models across merging methods, reported for fully fine-tuned models and for LoRA-compressed models at ranks $r \in \{16, 32, 64, 128\}$. Under LoRA compression the gap between the individual-model reference and every merged variant widens sharply and monotonically as the rank decreases, exposing the catastrophic degradation that motivates our approach.

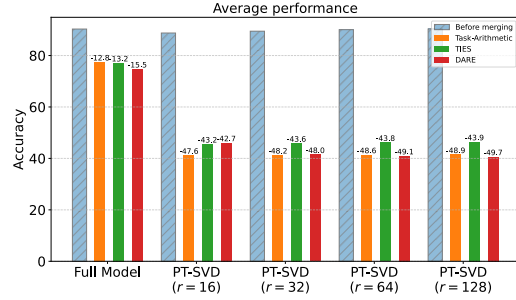


Figure 2: Average task performance of individual models (before merging) contrasted with merged models across merging methods, reported for fully fine-tuned models and for PT-SVD-compressed models at ranks $r \in \{16, 32, 64, 128\}$. As with LoRA, conventional merging collapses under low-rank compression while the individual-model reference remains high across all ranks, confirming that the failure is intrinsic to the compressed regime rather than an artifact of any particular compression scheme.

thus reduces to the problem of combining low-rank updates across tasks—that is, constructing a merged model that integrates the compressed deltas $\hat{\Delta}_i = \{\hat{\Delta}_i^l\}_{l=1}^L$ over all tasks:

$$\hat{\theta}_{\mathcal{M}} = \theta_{\text{pre}} + \mathcal{M}(\{\hat{\Delta}_i\}_{i=1}^n), \quad \max_{\mathcal{M}} \frac{1}{n} \sum_{i=1}^n \text{Perf}(\hat{\theta}_{\mathcal{M}}, \mathcal{D}_i), \quad (2)$$

where L is the total number of layers. Throughout this work we concentrate on this setting, in which every model is stored in low-rank form. The next section investigates, both empirically and mechanistically, how conventional merging strategies behave under this low-rank regime and establishes precisely why they fail.

4 MOTIVATING OBSERVATION

We now examine how conventional merging strategies behave when applied to the low-rank compressed models introduced above. For each layer l , prior work has considered two natural extensions of merging to the factorized setting (Tang et al., 2025; Zhao et al., 2025; Zheng et al., 2025; Zhang et al., 2025), which differ in whether the low-rank factors are recomposed before or after the merge:

- **Combined merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l \mathbf{B}_i^l\}_{i=1}^n),$$

in which each low-rank update is first expanded to its full-rank matrix form and only then merged;

- **Separate merging:**

$$\hat{\theta}_{\mathcal{M}}^l = \theta_{\text{pre}}^l + \mathcal{M}(\{\mathbf{A}_i^l\}_{i=1}^n) \mathcal{M}(\{\mathbf{B}_i^l\}_{i=1}^n),$$

in which the two low-rank factors are merged independently and then recomposed.

Although both formulations are mathematically valid, the combined strategy is untenable in practice on storage grounds. Retaining the merged update in combined form requires a full $m \times d$ matrix, whereas the separate strategy stores only the low-rank factors of sizes $\mathbf{A}_i^l \in \mathbb{R}^{m \times r}$ and $\mathbf{B}_i^l \in \mathbb{R}^{r \times d}$. Because $r(m+d) \ll md$ whenever $r \ll \min(m, d)$, the combined approach reintroduces precisely the overhead that compression was designed to eliminate. Concretely, for a layer with $m = d = 1024$ and rank $r = 16$, the combined form stores a full update of $md = 1,048,576$ parameters, whereas the separate form stores only $r(m+d) = 16 \times (1024 + 1024) = 32,768$ parameters—a

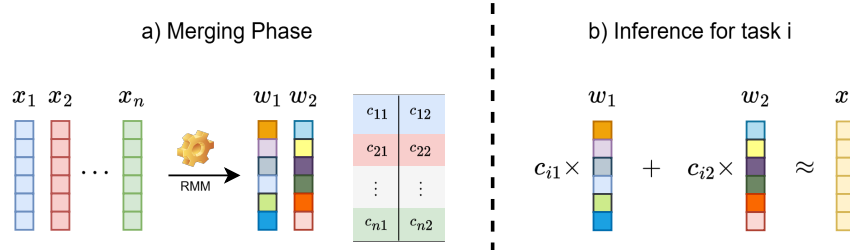


Figure 3: Overview of the proposed RMM framework. **Left:** in the merging phase, RMM computes a compact set of orthonormal basis directions (two, in this illustration) together with the task-specific coefficients that reconstruct each model from them, all in closed form and without task data. **Right:** at inference, target model i is reconstructed by linearly combining the shared basis vectors with its own coefficients and re-centering by the mean, thereby recovering the individual adapter on demand.

32× reduction in storage. Since the combined approach thereby forfeits the entire benefit of low-rank compression, we adopt the separate strategy as the only practically viable option for merging compressed models and employ it throughout the analyses that follow.

Using this separate strategy, we applied several established merging baselines to low-rank compressed models across a range of compression ratios. We adopted RoBERTa-base (Zhuang et al., 2021) as the pretrained backbone and evaluated on five GLUE tasks (Wang et al., 2018); full experimental details are provided in Appendix A.3. As Figures 1 and 2 make plain, merging compressed models induces a *catastrophic drop* in average task performance relative to merging their full-rank counterparts, and this drop is observed uniformly across every merging baseline we tested. We attribute the degradation to two mechanisms that act in concert: (i) **Amplified task interference:** compression discards much of the representational capacity, which leaves low-rank models markedly more susceptible to destructive interference when their updates are superimposed; (ii) **Broken coupling between A and B :** conventional merging treats the low-rank factors A_i^l and B_i^l as independent objects, disregards that only their product $A_i^l B_i^l$ carries meaning, and thereby injects structural misalignment into the merged updates. Acting in combination, these mechanisms severely impair merged low-rank models and reveal a deeper incompatibility between traditional merging and compressed representations—one that no amount of coefficient tuning within the single-model formulation can resolve, because it originates in the formulation itself rather than in the choice of coefficients.

This diagnosis yields a decisive conclusion: under conventional strategies, no single merged model can reliably recover the task-specific performance of the original low-rank adapters. It also points, however, to a constructive alternative: *rather than forcing all updates into one representation, can we reconstruct the original compressed models—or closely approximate them—by retaining a compact set of basis models whose cost is equivalent to only a few adapters and can be tuned to the available storage budget?* Pursuing this question leads to an alternative design space in which merging produces a *reconstruction-capable model space* rather than a single collapsed model. We formalize this perspective in the next section and introduce our method, *Reversible Model Merging* (RMM).

5 REVERSIBLE MODEL MERGING

The empirical failures documented above expose a fundamental limitation: no single merged model can faithfully preserve the task-specific performance of all low-rank compressed models simultaneously. RMM responds by relinquishing the goal of collapsing all models into one set of weights. In its place, RMM constructs a small set of shared components—basis models—from which every original task model can be accurately reconstructed, thereby converting the merging problem from lossy consolidation into an invertible representation.

Rather than compressing all n task-specific models into a single merged solution $\hat{\theta}_{\mathcal{M}}$, RMM retains a compact set of p merged components with $2 \leq p \ll n$. These components span a low-dimensional subspace that captures the essential variation across tasks and thereby enables each model to be

recovered efficiently on demand. The formulation renders the performance–memory trade-off explicit and directly controllable: by tuning the single hyperparameter p , a practitioner selects any operating point along the continuum between aggressive consolidation (p small) and near-lossless reconstruction (p large).

For each layer l , we operate on the r -dimensional vectors that constitute the low-rank factors—the rows of A_i^l and the columns of B_i^l , where r is the compression rank. We designate each such row or column an independent *task vector* and its row or column index a *task vector position*. For a fixed position, we assemble the corresponding task vectors across all n models and denote the vector contributed by model i at that position by $\mathbf{x}_i \in \mathbb{R}^r$. RMM is then applied independently at each position: it compresses the set $\{\mathbf{x}_i\}_{i=1}^n$ by projecting these vectors onto a shared basis from which they can subsequently be reconstructed. Since each layer contributes $m + d$ positions in total— m rows from A_i^l and d columns from B_i^l —the identical procedure is executed separately at every one of them. Crucially, because it operates directly on the factors, reconstruction never expands the update to its full-rank $m \times d$ form and so preserves the storage savings of compression at every step.

Concretely, at each position we seek a set of basis vectors $\{\mathbf{w}_1, \dots, \mathbf{w}_p\}$ with each $\mathbf{w}_j \in \mathbb{R}^r$ for $1 \leq j \leq p$. Their span defines a subspace $\mathbf{W} = [\mathbf{w}_1 \dots \mathbf{w}_p] \in \mathbb{R}^{r \times p}$ onto which each task vector $\mathbf{x}_i$ is projected, subject to the requirement that the projection reconstruct $\mathbf{x}_i$ as accurately as possible. We further introduce the coefficient matrix $\mathbf{C} \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ holds the scalar weights that linearly combine the basis vectors to reconstruct $\mathbf{x}_i$. The design objective is to determine the pair $(\mathbf{W}, \mathbf{C})$ that jointly minimizes the aggregate squared reconstruction error over all n task vectors, which we formalize as

$$\min_{\mathbf{W}, \mathbf{C}} \sum_{i=1}^n \left\| \mathbf{x}_i - \sum_{j=1}^p c_{ij} \mathbf{w}_j \right\|_2^2. \quad (3)$$

This representation stores rp parameters for the basis and np for the coefficients, $p(r+n)$ in total—a substantial reduction from the rn parameters needed to retain all task vectors directly whenever $p \ll \min(r, n)$. The hyperparameter p therefore governs the performance–storage trade-off directly; Figure 3 illustrates the construction for $p = 2$. Stacking the task vectors row-wise, the objective in equation 3 admits the compact matrix form

$$\min_{\mathbf{W}, \mathbf{C}} \left\| \mathbf{X} - \mathbf{C} \mathbf{W}^\top \right\|_F^2, \quad (4)$$

where $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$ stacks all task vectors as rows. For any fixed $\mathbf{W}$, the inner minimization over $\mathbf{C}$ is an ordinary least-squares problem that admits the closed-form solution $\mathbf{C}^* = \mathbf{X} \mathbf{W} (\mathbf{W}^\top \mathbf{W})^{-1}$. Constraining the basis to be orthonormal ($\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$ with $p < r$) collapses this expression to $\mathbf{C}^* = \mathbf{X} \mathbf{W}$ and reduces the joint objective to the single-variable program

$$\min_{\mathbf{W}} \left\| \mathbf{X} - \mathbf{X} \mathbf{W} \mathbf{W}^\top \right\|_F^2 \quad \text{s.t.} \quad \mathbf{W}^\top \mathbf{W} = \mathbf{I}_p. \quad (5)$$

Under this formulation we store the basis $\mathbf{W} \in \mathbb{R}^{r \times p}$ together with $\mathbf{X} \mathbf{W} \in \mathbb{R}^{n \times p}$, the latter being precisely the coefficient matrix $\mathbf{C}$ of equation 4.

Theorem 1 (Optimal Solution to equation 5). *Let $\mathbf{X} \in \mathbb{R}^{n \times r}$ be zero-centered and $p < \min\{r, n\}$. The optimal basis $\mathbf{W}^*$ that minimizes equation 5 is given by the top- p eigenvectors of the sample covariance matrix $\mathbf{X}^\top \mathbf{X}$. Equivalently, $\mathbf{W}^*$ consists of the top- p right singular vectors of $\mathbf{X}$.*

The proof of Theorem 1 is provided in Appendix A.1. This closed-form characterization is the crux of RMM: it renders basis selection simultaneously data-free and optimal in the reconstruction sense, and thereby obviates the iterative optimization and data access that competing methods require. We now describe the two phases of RMM that operationalize this result.

5.1 MERGING PHASE

The merging phase constructs the compact shared-basis representation of all task-specific models and is performed offline, prior to any inference. For each layer and each task vector position (a fixed row of A_i^l or column of B_i^l), we assemble the task vectors of all n models into $\mathbf{X} = [\mathbf{x}_1 \dots \mathbf{x}_n]^\top \in \mathbb{R}^{n \times r}$. We then compute the mean vector $\boldsymbol{\mu} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ and zero-center the

data ($\mathbf{X} \leftarrow [\mathbf{x}_1 - \boldsymbol{\mu} \dots \mathbf{x}_n - \boldsymbol{\mu}]^\top$), a step that fulfills the zero-centering precondition of Theorem 1. We next factorize the centered matrix by SVD, $\mathbf{X} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$, and take its top- p right singular vectors as the optimal orthonormal basis for equation 5, $\mathbf{W}^* = \mathbf{V}_{[:,1:p]} \in \mathbb{R}^{r \times p}$. Finally, we form the coefficient matrix $\mathbf{C}^* = \mathbf{X}\mathbf{W}^* \in \mathbb{R}^{n \times p}$, whose i -th row $[c_{i1}, \dots, c_{ip}]$ supplies precisely the weights that reconstruct $\mathbf{x}_i$ from the shared basis. Because Theorem 1 certifies this basis as globally optimal for the reconstruction objective, the merging phase requires neither iterative refinement nor any access to task data.

On completing this phase we discard the original task vectors $\{\mathbf{x}_i\}_{i=1}^n$ and retain, per position, only the basis $\mathbf{W}^*$, the coefficients $\mathbf{C}^*$, and the mean vector $\boldsymbol{\mu}$. This reduces the per-position footprint from the rn parameters needed to store all task vectors to $p(r+n)+r$, where the additive r accounts for the retained mean. Since the phase runs entirely offline and without task data, its cost is incurred exactly once and leaves inference-time efficiency wholly unaffected.

5.2 INFERENCE PHASE

Consistent with common practice in model merging, we assume that the target task for each input is either specified by the user or resolved by an oracle router that maps the input to the most suitable model index. Given the target index i , the inference phase reconstructs its task vector $\mathbf{x}_i$ as $\hat{\mathbf{x}}_i$ from the compact per-position representation $(\mathbf{W}^*, \mathbf{C}^*, \boldsymbol{\mu})$ produced during merging. Reassembling the reconstructed vectors $\hat{\mathbf{x}}_i$ across all positions—the rows for $\mathbf{A}_i^l$ and the columns for $\mathbf{B}_i^l$ —recovers the complete low-rank factors of model i .

Recall that, at each position, $\mathbf{W}^* \in \mathbb{R}^{r \times p}$ holds the orthonormal basis, $\mathbf{C}^* \in \mathbb{R}^{n \times p}$ the task-specific coefficients, and $\boldsymbol{\mu} \in \mathbb{R}^r$ the mean across task vectors. Reconstruction then reduces to a linear combination of the basis directions weighted by model i ’s own coefficients and re-centered by the mean $\boldsymbol{\mu}$:

$$\hat{\mathbf{x}}_i = \sum_{j=1}^p c_{ij}^* \mathbf{w}_j^* + \boldsymbol{\mu} = \mathbf{C}_{[i,:]}^* \mathbf{W}^{*\top} + \boldsymbol{\mu}.$$

The resulting $\hat{\mathbf{x}}_i \in \mathbb{R}^r$ approximates the original task vector $\mathbf{x}_i$ discarded during merging, with the approximation error controlled monotonically by the choice of p : larger p retains more of the shared subspace and reduces reconstruction error toward zero. Executed independently at every position and layer, this reconstruction yields the complete low-rank delta $\hat{\Delta}_i$ of task-specific model i . Algorithm 1 states the full merging and inference procedures in detail. We next evaluate RMM empirically and benchmark it against standard baselines.

6 EXPERIMENTS

To assess RMM comprehensively, we conduct extensive experiments spanning multiple compression settings, task suites, and model architectures. Our evaluation is designed to answer two questions: whether RMM preserves the performance of low-rank task-specific models while reducing overall storage, and how it compares to conventional merging baselines across these regimes. As the results below establish, RMM answers both affirmatively and by a wide, consistent margin.

6.1 EXPERIMENTAL SETUP

We first evaluate RMM on eight RoBERTa-base models fine-tuned for eight distinct tasks (Zhuang et al., 2021). Each model is compressed independently with either PT-SVD or LoRA at target ranks $r \in \{16, 32, 64, 128\}$, after which RMM is applied to the resulting low-rank models and compared against three standard merging baselines: Task Arithmetic (TA) (Ilharco et al., 2023), TIES-merging (Yadav et al., 2023), and DARE (Yu et al., 2024). As an upper bound we additionally report the performance of the individual compressed models before merging, a reference that no merging method can exceed by construction. To probe generality across backbones, modalities, and task counts, Appendix A.2 reports further experiments on the decoder-only OPT-1.3b (Zhang et al., 2022) and on the vision transformer ViT-B/32 (Radford et al., 2021) for image classification tasks, so that the evaluation collectively covers encoder, decoder, and vision-transformer architectures across both language and vision modalities.

Algorithm 1 Reversible Model Merging (RMM)

```
1: Merging Phase:
2: Require: Low-rank deltas  $A_i^l$  and  $B_i^l$ , compression rank  $r$ , number of basis vectors  $p$ 
3: for  $l = 1$  to  $L$  do
4:   for each task vector position (each row in  $A_i^l$  or each column in  $B_i^l$ ) do
5:     Gather task vectors  $x_i$  of each model  $i$  ( $1 \leq i \leq n$ )
6:      $X \leftarrow [x_1 \dots x_n]^\top \in \mathbb{R}^{n \times r}$ 
7:      $\mu \leftarrow \sum_{i=1}^n x_i \in \mathbb{R}^r$ 
8:      $X \leftarrow [x_1 - \mu \dots x_n - \mu]^\top$ 
9:      $U, \Sigma, V^\top \leftarrow \text{SVD}(X)$ 
10:     $W^* \leftarrow V_{[:,1:p]} \in \mathbb{R}^{r \times p}$ 
11:     $C^* \leftarrow XW^* \in \mathbb{R}^{n \times p}$ 
12:    Store  $(W^*, C^*, \mu)$  and discard task vectors  $\{x_1 \dots x_n\}$ 
13:   end for
14: end for
15: Inference Phase:
16: Require: target task  $i$ , stored representation  $(W^*, C^*, \mu)$  for each task vector position
17: for  $l = 1$  to  $L$  do
18:   for each task vector position do
19:      $\hat{x}_i = C_{[i,:]}^* W^{*\top} + \mu$ 
20:   end for
21:    $\hat{A}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all row positions})$ 
22:    $\hat{B}_i^l = \text{Concatenate}(\hat{x}_i \text{ vectors across all column positions})$ 
23: end for
24:  $\hat{\Delta}_i = \{\hat{A}_i^l \hat{B}_i^l\}_{l=1}^L$ 
25: Inference with the reconstructed model  $\theta_{pre} + \hat{\Delta}_i$ 
```

We follow the GLUE benchmark protocol (Wang et al., 2018) and evaluate each task with its standard metric: QNLI (question–answering inference), MRPC (paraphrase detection), SST-2 (sentiment classification), MNLI (multi-genre natural language inference), QQP (duplicate question detection), and RTE (textual entailment) by accuracy; CoLA (linguistic acceptability) by Matthews correlation; and STS-B (semantic textual similarity) by Pearson correlation. The average across tasks serves as the overall performance measure. Alongside accuracy, we quantify the relative storage cost of each method as the fraction of parameters it retains relative to storing all individual task-specific adapters. Baselines that store a single merged model incur a fixed cost of $\frac{r}{rn} = \frac{1}{n}$, whereas RMM with p basis components incurs $\frac{p(r+n)+r}{rn}$, as derived in Section 5.1. Reporting both axes together renders the accuracy–storage trade-off directly legible in the tables that follow and situates every accuracy gain against its precise storage cost, so that RMM’s advantage can be assessed on both dimensions simultaneously.

6.2 RESULTS AND DISCUSSION

Tables 1 and 2 report RMM against the baselines under PT-SVD and LoRA compression, respectively. Across both compression schemes and all four target ranks $r \in \{16, 32, 64, 128\}$, RMM outperforms TA, TIES, and DARE by a wide and uniform margin, without a single exception. At the lower ranks ($r = 16, 32$) the single-model baselines fail catastrophically, attaining only 31–42% average score and collapsing almost entirely on the structurally sensitive tasks CoLA and STS-B. RMM with $p = 2$ already recovers a substantial share of the lost performance (57.24% under PT-SVD and 51.41% under LoRA at $r = 16$), and RMM with $p = 3$ narrows the gap to the individual compressed models still further (reaching 72.22% and 58.64% at $r = 16$). At the higher ranks ($r = 64, 128$), where the compression error is smaller, RMM continues to dominate: with $p = 3$ it attains 64.34% and 60.58% under PT-SVD and 55.40% and 61.35% under LoRA, while the single-model baselines remain stalled near 40%. Even the more economical $p = 2$ configuration retains a clear advantage throughout. Taken together, these patterns establish that the reversible formulation structurally averts the destructive interference that cripples prior methods rather than merely attenuating it: because each task is reconstructed from its own coefficients rather than sharing one

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	77.98	58.05	90.59	83.80
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.19	31.66
	TIES	13%	50.56	34.78	49.08	31.82	36.82	47.29	0.00	3.75	31.76
	DARE	13%	50.80	33.51	49.08	31.82	36.82	47.29	0.00	5.07	31.80
	RMM ($p = 2$)	50%	51.20	78.20	88.30	66.74	86.02	47.29	6.56	33.59	57.24
	RMM ($p = 3$)	69%	88.83	86.26	92.89	79.58	87.18	48.74	33.85	60.46	72.22
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	78.34	59.33	90.68	84.46
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.21	31.66
	TIES	13%	50.56	35.19	49.08	31.82	36.82	47.29	0.00	3.87	31.83
	DARE	13%	50.72	33.51	49.08	31.82	36.82	47.29	0.00	3.48	31.59
	RMM ($p = 2$)	44%	50.54	74.32	55.05	62.68	87.07	47.29	0.00	23.18	50.02
	RMM ($p = 3$)	59%	84.66	86.32	92.89	79.70	88.05	47.29	23.57	44.54	68.38
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	78.70	61.11	90.72	85.10
	TA	13%	50.61	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.13	49.08	31.82	36.82	47.29	0.00	3.80	31.82
	DARE	13%	50.58	33.51	49.08	31.82	36.82	47.29	0.00	5.22	31.79
	RMM ($p = 2$)	41%	50.53	73.10	49.08	58.48	86.98	47.29	0.00	17.50	47.87
	RMM ($p = 3$)	55%	74.14	86.20	92.43	78.83	89.00	47.29	13.16	33.70	64.34
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	79.78	60.16	90.74	85.30
	TA	13%	50.59	33.51	49.08	31.82	36.82	47.29	0.00	4.23	31.67
	TIES	13%	50.58	35.54	49.08	31.82	36.82	47.29	0.00	3.93	31.88
	DARE	13%	50.61	34.78	49.08	31.82	36.82	47.29	0.00	2.41	31.60
	RMM ($p = 2$)	39%	50.54	70.78	49.08	54.69	84.68	47.29	0.00	14.53	46.45
	RMM ($p = 3$)	52%	59.20	85.80	92.32	77.50	89.02	47.29	6.56	26.94	60.58

Table 1: GLUE performance for merging eight RoBERTa-base models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$. *No merging* denotes the individual-model upper bound; TA, TIES, and DARE store a single merged model, whereas RMM stores p reversible components. At every rank RMM ($p=3$) improves the average score over the strongest single-model baseline by a decisive margin—from roughly 40 points at $r=16$ (72.22 vs. 31.80) to roughly 29 points at $r=128$ (60.58 vs. 31.88)—and RMM ($p=2$) already surpasses every baseline, at the higher storage fractions reported in the Storage column. The advantage is most pronounced on CoLA and STS-B, the structurally sensitive tasks on which all single-model baselines collapse to near-zero while RMM retains substantial accuracy.

collapsed weight vector, the interference channel that governs single-model merging is eliminated by construction rather than mitigated by tuning.

We further observe that the relative gain of RMM over the baselines narrows as the rank grows. A natural explanation is that higher-rank task representations occupy higher-dimensional, more expressive, and less mutually aligned subspaces, which are inherently harder to compress into a shared low-dimensional basis; this effect is expected and, crucially, never reverses the ranking, since RMM remains ahead at every rank. Taken together, the results establish a consistent and uniform advantage for RMM across both compression regimes and all ranks: it preserves high task performance in exchange for additional storage, and the number of reversible components p exposes this accuracy-storage trade-off as an explicit, tunable dial. Crucially, the added cost is confined to **off-line memory**—typically inexpensive, CPU-resident storage—whereas task performance governs runtime quality. In such settings, a modest increase in CPU-side storage is a decidedly favorable exchange for markedly higher accuracy, and it is precisely the exchange that the single-model baselines are structurally unable to offer.

6.3 SCALABILITY OF RMM

We now examine how RMM scales as the number of models n grows. Figures 4 and 5 plot the storage ratio, relative to retaining all individual task-specific models, against the number of tasks for $p = 2$ and $p = 3$, respectively. Both curves confirm favorable scaling: the relative storage cost decreases markedly as tasks accumulate. This behavior is the central operational benefit of reversible merging—storage grows sublinearly in the number of task-specific models—and it follows directly from the design. Once the shared basis is fixed, incorporating a new task appends only a single row to the coefficient matrix, whereas conventional approaches face a stark dichotomy: either commit to one merged model and forfeit per-task performance, or store every task-specific model independently and incur strictly linear growth. RMM is therefore particularly well suited to large-scale

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	RTE	COLA	STS-B	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	68.23	53.92	88.71	77.71
	TA	13%	50.59	33.51	72.13	34.79	64.46	47.29	0.00	21.81	40.57
	TIES	13%	50.63	33.68	74.31	35.64	67.45	47.29	0.00	26.38	41.92
	DARE	13%	50.63	33.51	73.74	35.97	65.35	47.29	0.00	21.62	41.01
	RMM ($p = 2$)	50%	51.13	34.14	90.71	60.48	84.41	47.29	0.00	43.14	51.41
	RMM ($p = 3$)	69%	55.92	38.14	93.12	82.95	87.30	47.29	6.56	57.81	58.64
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	62.82	53.12	88.53	76.54
	TA	13%	50.91	68.06	54.70	33.77	65.50	55.96	0.00	-0.48	41.05
	TIES	13%	51.03	68.06	56.77	34.24	66.27	58.12	0.00	1.02	41.94
	DARE	13%	50.96	64.64	63.07	35.12	63.66	55.60	0.00	1.01	41.76
	RMM ($p = 2$)	44%	52.08	65.39	81.54	62.46	77.06	57.40	0.00	9.95	50.73
	RMM ($p = 3$)	59%	54.62	69.97	92.20	81.29	85.97	57.40	4.64	24.08	58.77
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	67.15	58.31	88.70	78.25
	TA	13%	54.75	68.52	77.29	37.24	36.82	47.29	0.00	-3.53	39.80
	TIES	13%	54.82	68.75	80.73	37.67	36.82	47.29	0.00	-2.48	40.45
	DARE	13%	53.71	68.58	78.56	38.33	36.82	47.29	0.00	-4.28	39.88
	RMM ($p = 2$)	41%	55.74	69.28	87.27	42.81	77.72	47.29	0.00	4.91	48.13
	RMM ($p = 3$)	55%	59.07	70.14	90.83	79.11	82.06	47.29	0.00	14.70	55.40
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	57.04	59.35	89.68	77.71
	TA	13%	51.49	69.22	54.47	37.16	63.27	47.29	0.00	-1.74	40.15
	TIES	13%	51.91	68.93	57.34	37.81	63.36	47.29	0.00	-0.25	40.80
	DARE	13%	51.97	68.99	58.60	35.64	63.33	47.29	0.00	-0.44	40.67
	RMM ($p = 2$)	39%	57.18	67.83	88.53	49.54	84.64	47.29	0.00	28.00	52.88
	RMM ($p = 3$)	52%	59.13	39.19	92.09	80.08	86.39	47.29	15.45	71.18	61.35

Table 2: GLUE performance for merging eight RoBERTa-base models compressed with LoRA at ranks $r \in \{16, 32, 64, 128\}$. Column conventions match Table 1. RMM ($p=3$) again leads at every rank, exceeding the strongest baseline average by roughly 15–21 points (for instance, 58.64 vs. 41.92 at $r=16$ and 61.35 vs. 40.80 at $r=128$), and RMM ($p=2$) remains ahead of all single-model baselines throughout. The advantage thus holds under LoRA compression as robustly as under PT-SVD, establishing that RMM’s superiority is invariant to the compression scheme.

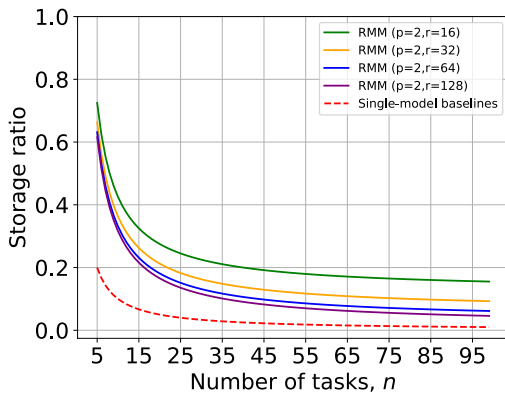


Figure 4: Scalability of RMM ($p = 2$): the storage ratio—the memory required, expressed as a percentage of storing all models—as a function of the number of tasks. The ratio falls steadily as tasks accumulate, direct evidence that storage grows sublinearly in the number of task-specific models.

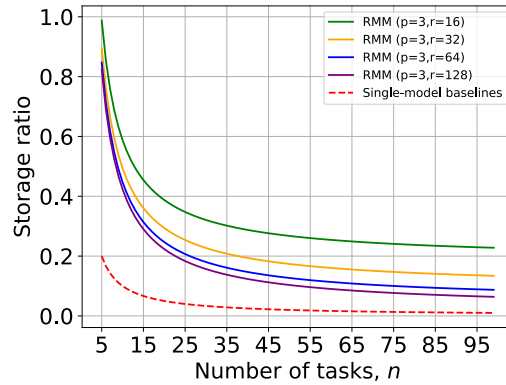


Figure 5: Scalability of RMM ($p = 3$): the storage ratio—the memory required, expressed as a percentage of storing all models—as a function of the number of tasks. As with $p = 2$, the ratio declines with the task count, so the higher reconstruction fidelity afforded by $p = 3$ is obtained without sacrificing the favorable sublinear scaling.

multi-task, continual, and federated deployments that must manage hundreds of models efficiently without compromising per-task performance, and its advantage becomes more, not less, pronounced precisely as the collection grows.

7 CONCLUSION

We introduced a principled framework for reversible model merging that redefines how multiple fine-tuned models are stored, merged, and reconstructed. Rather than forcing all task-specific knowledge into a single merged model, RMM relaxes this constraint by retaining a compact set of shared components—substantially fewer than the number of original models, yet strictly more expressive than a single aggregate—from which each model can be recovered by linear combination. Grounded in a closed-form, data-free basis-selection solution that Theorem 1 certifies as optimal in the reconstruction sense, this design reconstructs task-specific models accurately while sharply reducing storage overhead, and it exposes a controllable accuracy–storage trade-off that is especially valuable for low-rank compressed models. By representing fine-tuned models through compact bases and reversible mappings, we establish that models can be merged, reconstructed, and deployed at high accuracy under strict memory budgets, with the degree of compression and the reconstruction fidelity tuned to the demands of the application—a capability we validate across encoder and decoder language models as well as vision transformers.

More broadly, by formalizing merging as a structured, reversible process, our framework moves decisively beyond heuristic, one-shot merging. This perspective not only enlarges the design space of efficient multi-task deployment but also charts concrete directions for future research on performance-aware compression and reconstruction.

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A APPENDIX

A.1 PROOF OF THEOREM 1

Let $\mathbf{A} = \mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top$ denote the reconstruction residual. By the definition of the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{Tr}(\mathbf{A}^\top \mathbf{A})$, where $\text{Tr}(\cdot)$ denotes the trace of a square matrix. Expanding the residual yields

$$\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 = \text{Tr}((\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)).$$

Expanding the argument of the trace gives

$$\begin{aligned} (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top)^\top (\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) &= (\mathbf{X}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top)(\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top) \\ &= \mathbf{X}^\top \mathbf{X} - \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top - \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X} \\ &\quad + \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}\mathbf{W}^\top. \end{aligned}$$

Taking the trace of this expansion and invoking the cyclic property of the trace together with the orthonormality constraint $\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p$, we obtain

$$\begin{aligned} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{W}\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - 2\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) + \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}) \\ &= \text{Tr}(\mathbf{X}^\top \mathbf{X}) - \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}). \end{aligned}$$

Because $\text{Tr}(\mathbf{X}^\top \mathbf{X})$ is independent of $\mathbf{W}$, minimizing $\|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2$ is equivalent to maximizing $\text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W})$. Consequently,

$$\min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \|\mathbf{X} - \mathbf{X}\mathbf{W}\mathbf{W}^\top\|_F^2 \iff \max_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}_p} \text{Tr}(\mathbf{W}^\top \mathbf{X}^\top \mathbf{X}\mathbf{W}).$$

By the Rayleigh quotient theorem (Horn & Johnson, 2012), this trace is maximized by taking the columns of $\mathbf{W}$ to be the top- p eigenvectors of $\mathbf{X}^\top \mathbf{X}$. This is precisely the principal component analysis (PCA) solution: it identifies the p -dimensional subspace that maximizes the projected variance and, equivalently, minimizes the squared reconstruction error. Moreover, writing the SVD of $\mathbf{X}$ as $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ yields $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{\Sigma}^2\mathbf{V}^\top$, so that the eigenvectors of $\mathbf{X}^\top \mathbf{X}$ coincide with the right singular vectors of $\mathbf{X}$ (the columns of $\mathbf{V}$). The optimal basis can therefore be read off directly from the top- p right singular vectors of $\mathbf{X}$ —an SVD-based route that is numerically more stable than, and frequently computationally preferable to, explicitly forming $\mathbf{X}^\top \mathbf{X}$, particularly when r is large. This is exactly the construction employed in the merging phase of Algorithm 1, and it establishes that RMM’s basis is optimal by construction rather than merely heuristic.

A.2 ADDITIONAL RESULTS

To establish the generality of RMM beyond RoBERTa, we present extended results for two additional pretrained backbones that span distinct architectures and modalities: OPT-1.3b and ViT-B/32. For **OPT-1.3b**, a decoder-only language model an order of magnitude larger than RoBERTa-base, we evaluate on six natural language understanding tasks drawn from GLUE and related benchmarks—QNLI, MRPC, SST-2, MNLI, QQP, and BoolQ (Clark et al., 2019). For **ViT-B/32**, we turn to vision classification across seven diverse datasets: MNIST (LeCun et al., 1998), GTSRB (Stallkamp et al., 2011), SVHN (Netzer et al., 2011), Cars (Krause et al., 2013), EuroSAT (Helber et al., 2019), DTD (Cimpoi et al., 2014), and SUN397 (Xiao et al., 2010). Together these backbones exercise RMM across encoder, decoder, and vision-transformer architectures and across both language and vision modalities, thereby subjecting the reversible formulation to a demanding test of cross-architecture and cross-modal generality.

Table 3 reports the six OPT-1.3b models compressed with PT-SVD. Both RMM variants consistently outperform the baselines, and RMM at $p = 3$ narrows the gap to the original individual models while requiring substantially fewer parameters, which confirms that the advantages observed on RoBERTa carry over intact to a much larger decoder-only language model. Table 4 reports the seven ViT-B/32 models, where the benefit of RMM is more pronounced still: at $p = 3$ it attains an average accuracy less than 3% below that of the original models while reducing storage to only 59% of the cost of retaining all models at $r = 128$. Taken together, these results establish that RMM faithfully reconstructs the original compressed models across architectures and modalities while delivering substantial storage efficiency, and they demonstrate that its effectiveness is a general property of the reversible formulation rather than an artifact of any single backbone or task family.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	BOOLQ	Average
16	No merging	100%	91.74	71.54	95.30	84.19	88.14	71.47	83.73
	Task-Arithmetic	17%	50.34	37.16	58.49	34.43	63.30	41.87	47.60
	TIES	17%	49.79	36.75	61.24	35.57	63.55	40.67	47.93
	DARE	17%	49.95	37.68	56.08	34.51	63.39	42.78	47.40
	RMM ($p = 2$)	63%	88.12	44.00	93.58	73.06	85.63	59.69	74.01
	RMM ($p = 3$)	85%	91.58	49.04	95.64	80.31	87.52	65.50	78.27
32	No merging	100%	92.06	72.17	95.41	84.60	89.17	72.45	84.31
	Task-Arithmetic	17%	50.28	37.22	58.94	34.57	63.33	41.93	47.71
	TIES	17%	49.77	36.93	61.93	35.69	63.65	40.73	48.12
	DARE	17%	50.39	40.87	56.19	34.96	63.59	43.33	48.22
	RMM ($p = 2$)	56%	86.66	41.91	91.74	71.75	85.68	56.39	72.36
	RMM ($p = 3$)	76%	91.85	44.81	94.84	80.21	88.43	63.52	77.28
64	No merging	100%	92.26	72.12	95.41	84.64	90.07	72.84	84.56
	Task-Arithmetic	17%	50.28	37.28	59.06	34.56	63.34	42.08	47.77
	TIES	17%	49.75	37.04	61.81	35.84	63.74	40.76	48.16
	DARE	17%	50.16	40.52	58.83	35.59	63.35	42.54	48.50
	RMM ($p = 2$)	53%	83.87	42.72	89.45	69.76	85.70	53.43	70.82
	RMM ($p = 3$)	71%	92.07	41.97	94.50	79.70	88.73	61.96	76.49
128	No merging	100%	92.35	72.41	95.41	84.67	90.93	73.09	84.81
	Task-Arithmetic	17%	50.32	37.39	58.94	34.63	63.37	42.08	47.79
	TIES	17%	49.79	37.04	62.04	35.99	63.74	40.55	48.19
	DARE	17%	50.03	38.14	57.91	34.72	63.31	42.11	47.70
	RMM ($p = 2$)	52%	81.38	42.78	84.75	67.66	83.22	51.99	68.63
	RMM ($p = 3$)	69%	92.28	41.80	93.69	79.42	88.76	59.69	75.94

Table 3: GLUE performance for merging six OPT-1.3b models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$. Both RMM variants surpass every single-model baseline at all ranks; with $p=3$, RMM recovers average scores that closely track the individual-model upper bound (78.27 vs. 83.73 at $r=16$) while retaining only a fraction of the parameters needed to store all adapters. The gains observed on RoBERTa thus transfer intact to a decoder-only backbone an order of magnitude larger, evidence that RMM’s advantage is not tied to model scale.

A.3 RESULTS FOR 5 TASKS

The motivating study of Section 4 adopted RoBERTa-base as the pretrained backbone and evaluated the merging baselines on five GLUE tasks (QNLI, MRPC, SST-2, MNLI, QQP); Figures 1 and 2 summarize those results at a glance. For completeness, we provide the underlying per-task numbers here: Table 5 accompanies Figure 1 and reports the LoRA fine-tuned models together with the improvements delivered by RMM, whereas Table 6 accompanies Figure 2 and reports the PT-SVD compressed models. These tables make explicit that, even in this reduced five-task regime, RMM recovers near-individual-model performance precisely where every baseline collapses.

r	Method	Storage	MNIST	GTSRB	SVHN	Cars	EuroSAT	DTD	SUN397	Average
16	No merging	100%	99.61	97.22	97.06	69.20	99.67	75.21	72.72	87.24
	Task-Arithmetic	14%	47.19	28.29	38.64	56.27	57.52	41.38	62.92	47.46
	TIES	14%	60.14	34.52	49.03	58.55	61.70	43.67	62.87	52.93
	DARE	14%	53.80	31.00	44.96	57.60	56.93	43.99	62.42	50.10
	RMM ($p = 2$)	55%	99.16	88.52	94.89	66.61	97.44	60.64	67.86	82.16
	RMM ($p = 3$)	76%	99.58	95.65	96.54	68.30	98.93	67.18	69.50	85.10
32	No merging	100%	99.65	98.45	97.35	73.04	99.78	78.72	75.52	88.93
	Task-Arithmetic	14%	47.90	28.31	38.68	56.24	57.52	41.38	63.00	47.58
	TIES	14%	61.25	35.28	49.95	58.80	62.41	44.04	63.05	53.54
	DARE	14%	53.05	31.04	46.34	58.54	57.74	42.87	62.87	50.35
	RMM ($p = 2$)	49%	98.98	88.23	94.29	68.32	96.81	61.91	68.36	82.41
	RMM ($p = 3$)	67%	99.50	96.53	96.68	70.90	98.93	70.43	71.23	86.31
64	No merging	100%	99.67	98.68	97.41	75.76	99.85	79.15	77.73	89.75
	Task-Arithmetic	14%	47.99	28.30	38.71	56.21	57.59	41.38	63.22	47.63
	TIES	14%	61.83	35.93	50.28	59.08	63.07	44.15	63.17	53.93
	DARE	14%	54.78	32.79	45.05	58.23	61.30	42.87	62.77	51.11
	RMM ($p = 2$)	46%	98.46	84.43	92.83	69.36	95.59	62.39	69.77	81.83
	RMM ($p = 3$)	62%	99.45	96.14	96.33	72.84	98.56	72.07	73.85	87.03
128	No merging	100%	99.68	98.75	97.46	76.99	99.85	79.04	78.59	90.05
	Task-Arithmetic	14%	48.00	28.31	38.73	56.21	57.63	41.44	63.35	47.67
	TIES	14%	62.08	36.36	50.48	59.23	63.41	44.68	63.25	54.21
	DARE	14%	55.16	32.07	45.47	58.55	61.85	43.09	62.85	51.29
	RMM ($p = 2$)	44%	97.22	78.56	90.63	70.22	94.00	61.49	71.03	80.45
	RMM ($p = 3$)	59%	99.28	94.59	95.95	74.24	98.19	71.86	76.02	87.16

Table 4: Vision performance for merging seven ViT-B/32 models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$. The reversible formulation is especially effective in the vision setting: RMM ($p=3$) tracks the individual-model upper bound closely across all ranks (87.16 vs. 90.05 at $r=128$), remaining within roughly three points while storing only 59% of the parameters, whereas the single-model baselines lose tens of points. RMM therefore extends beyond language to a distinct architecture and modality without any modification, confirming that its effectiveness is a general property of the reversible formulation.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	61.38	83.48	94.50	83.90	87.56	82.16
	Task-Arithmetic	20%	50.61	33.51	73.85	34.23	64.58	51.36
	TIES	20%	50.65	33.51	75.57	35.83	66.97	52.51
	DARE	20%	50.61	33.51	79.36	34.70	63.40	52.32
	RMM ($p = 2$)	73%	59.64	68.64	93.00	82.68	86.89	78.17
32	No merging	100%	60.13	81.28	93.81	84.46	88.18	81.57
	Task-Arithmetic	20%	51.18	67.30	58.83	34.33	68.56	56.04
	TIES	20%	51.36	68.17	59.52	34.76	71.46	57.05
	DARE	20%	50.83	64.87	68.12	35.54	68.93	57.66
	RMM ($p = 2$)	66%	57.72	67.54	92.55	81.80	86.13	77.15
64	No merging	100%	59.62	84.12	94.15	85.20	88.75	82.37
	Task-Arithmetic	20%	54.74	68.93	80.28	35.70	36.82	55.29
	TIES	20%	55.26	68.93	78.78	36.76	36.82	55.31
	DARE	20%	54.71	68.81	74.54	38.72	36.82	54.72
	RMM ($p = 2$)	63%	59.82	67.48	91.51	83.21	84.41	77.29
128	No merging	100%	61.58	85.04	93.58	85.71	89.68	83.12
	Task-Arithmetic	20%	52.52	68.70	54.82	37.83	63.46	55.47
	TIES	20%	52.55	69.04	57.80	38.91	63.72	56.40
	DARE	20%	55.04	69.62	54.47	38.92	64.12	56.43
	RMM ($p = 2$)	62%	60.21	34.96	91.86	78.59	86.65	70.45

Table 5: GLUE performance for merging five RoBERTa-base models compressed with LoRA at ranks $r \in \{16, 32, 64, 128\}$ (per-task detail for Figure 1). RMM ($p=2$) lifts the average score far above every baseline—78.17 against a best baseline of 52.51 at $r=16$ —and approaches the individual-model upper bound at every rank, while the baselines never exceed the upper 50s.

r	Method	Storage	QNLI	MRPC	SST-2	MNLI	QQP	Average
16	No merging	100%	92.02	85.68	93.69	85.21	87.19	88.76
	Task-Arithmetic	20%	53.03	35.19	49.08	31.82	36.82	41.19
	TIES	20%	55.46	54.32	49.08	31.82	37.21	45.58
	DARE	20%	53.69	58.67	49.08	31.82	37.08	46.07
	RMM ($p = 2$)	73%	79.72	85.16	90.83	76.58	86.09	83.68
32	No merging	100%	92.31	85.86	94.15	86.38	88.62	89.46
	Task-Arithmetic	20%	53.23	35.59	49.08	31.82	36.82	41.31
	TIES	20%	55.78	55.19	49.08	31.82	37.62	45.90
	DARE	20%	55.89	33.74	49.20	31.82	36.85	41.50
	RMM ($p = 2$)	66%	63.87	82.78	91.28	77.12	87.05	80.42
64	No merging	100%	92.51	86.03	94.38	87.18	90.15	90.05
	Task-Arithmetic	20%	53.49	35.88	49.08	31.82	36.82	41.42
	TIES	20%	56.14	55.94	49.08	31.82	38.08	46.21
	DARE	20%	53.51	33.68	49.08	31.82	36.82	40.98
	RMM ($p = 2$)	63%	56.12	81.22	91.06	76.74	88.07	78.64
128	No merging	100%	92.68	86.14	94.50	87.33	91.05	90.34
	Task-Arithmetic	20%	53.60	36.06	49.08	31.82	36.82	41.48
	TIES	20%	56.40	56.35	49.08	31.82	38.61	46.45
	DARE	20%	50.52	34.96	49.08	31.82	36.92	40.66
	RMM ($p = 2$)	62%	54.11	79.19	90.60	74.49	87.46	77.17

Table 6: GLUE performance for merging five RoBERTa-base models compressed with PT-SVD at ranks $r \in \{16, 32, 64, 128\}$ (per-task detail for Figure 2). The pattern mirrors the LoRA case: RMM ($p=2$) reaches 83.68 at $r=16$ against a best baseline of 46.07 and remains close to the individual-model upper bound across all ranks, whereas the baselines languish in the low-to-mid 40s—confirming that the five-task advantage is likewise invariant to the compression scheme.